

Chunk Tier Verification

Evaluating positive identification and negative pool quality

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Overview

This notebook verifies the known act identification system that supports C1 (Measure ID) codebook development. The system classifies document chunks (10-page sliding windows) from US government documents into three tiers:

- **Tier 1:** Verbatim passage matches from `aligned_data` (gold standard)
- **Tier 2:** Act name matches via whitespace-normalized substring, subcomponent decomposition, or co-occurrence rules
- **Negative:** All remaining chunks, tagged with continuous `key_density` metric

We evaluate coverage, matching mechanisms, temporal consistency, and negative pool quality to ensure the tier system provides a robust foundation for LOOCV evaluation in Stage 2.

The identification of *known* acts is only implemented in the US, as it is the only country for which we know the acts from Romer and Romer (2010).

Corpus year filter

The evaluation corpus is filtered to documents published through 2007 (`max_doc_year = 2007`). Romer and Romer (2010) was published in 2010, with the last identified act signed in 2003 (Jobs and Growth Tax Relief Reconciliation Act). Documents through approximately 2007 represent the universe R&R had access to when conducting their analysis. Post-2007 documents can only contribute retrospective mentions of these acts, which would

inflate Tier 2 recall beyond what the actual identification task requires. Setting `max_doc_year = NULL` restores the full 1946-2022 corpus for sensitivity analysis.

Tier Distribution

Table 1: Distribution of chunks across three tiers

Tier	Count	Percentage
Tier 1	68	0.6%
Tier 2	1,511	12.2%
Negative	11,157	90.4%
Total	12,345	100.0%

The tier system identifies 68 Tier 1 chunks (verbatim passage matches) and 1,511 Tier 2 chunks (act name matches). The negative pool comprises 90.4% of all chunks, providing substantial diversity for hard-negative sampling in C1 evaluation.

Coverage Check

Table 2: Act coverage across tiers

Coverage Status	Acts	Percentage
Both tiers	30	76.9%
Tier 2 only	9	23.1%

Coverage Status: PASS

All 39 acts should have at least one tier match. All acts successfully matched.

Per-Act Chunk Matches

Table 3: Chunk matches per act across tiers

Act Name	Tier 1	Tier 2	Total
Economic Recovery Tax Act of 1981	3	94	97
Changes in Depreciation Guidelines and Revenue Act of 1962	2	84	86
Revenue Act of 1978	4	81	85
Tax Reform Act of 1976	1	82	83
Tax Reform Act of 1969	7	71	78
Revenue Act of 1950	1	70	71
Revenue Act of 1964	4	64	68
Revenue Act of 1971	2	59	61
Tax Reform Act of 1986	0	60	60
Tax Reduction Act of 1975	3	56	59
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	4	53	57
Social Security Amendments of 1983	0	53	53
Revenue Act of 1945	5	47	52
Omnibus Budget Reconciliation Act of 1990	0	47	47
Tax Equity and Fiscal Responsibility Act of 1982	0	46	46
Revenue Act of 1948	1	43	44
Tax Adjustment Act of 1966	2	38	40
Deficit Reduction Act of 1984	5	34	39
Federal-Aid Highway Act of 1956	1	38	39
Crude Oil Windfall Profit Tax Act of 1980	2	35	37
Excess Profits Tax Act of 1950	2	35	37
Revenue and Expenditure Control Act of 1968	2	34	36
Tax Reduction and Simplification Act of 1959	1	35	36
Revenue Act of 1959	0	30	30
Revenue Act of 1960	0	20	20
Revenue Act of 1961	0	17	17
Revenue Act of 1962	0	17	17
Revenue Act of 1963	0	17	17
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Revenue Act of 2026	0	17	17
Revenue Act of 2027	0	17	17
Revenue Act of 2028	0	17	17
Revenue Act of 2029	0	17	17
Revenue Act of 2030	0	17	17

Acts with zero matches indicate either: (1) no relevant passages in `aligned_data`, or (2) matching logic failures. High Tier 2 counts relative to Tier 1 suggest extensive discussion beyond labeled passages, which is expected for major legislation.

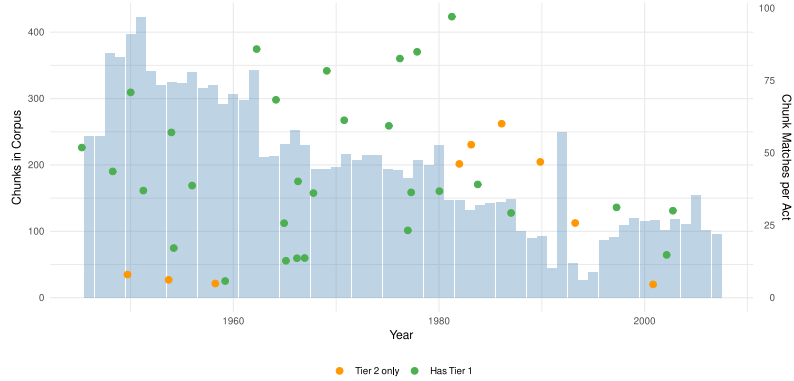


Figure 1: Corpus chunk density (bars, left axis) and total chunk matches per act (points, right axis) by year

The bars show corpus density over time (filtered to documents through 2007); the points overlay each act at its signed year with match count on the right axis. Green points have at least one verbatim Tier 1 match; orange points rely entirely on Tier 2 name matching. Acts in sparse corpus years tend to have fewer matches.

Matching Mechanism Breakdown

Table 4: Search terms and match counts by mechanism for each act

[illegible]

Table 5: Summary of matching mechanisms per act

Act	Year	Action		Split 'and'	Before		Par- enthet- ical	Par- enthet- ical + ar- ti- cle	to- tal
		+ sub- ject name	Full en		Co- occurrence	par- enthet- ical			
Eco- nomic Recoy- ery Tax Act of 1981 Changes in De- pre- ci- ation Guide- lines and Rev- enue Act of 1962 Rev- enue Act of 1978 Tax Reform Act- 1993-76 Public Law 101-508 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037 2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052 2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067 2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082 2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097 2098 2099 2100	1981	0	98	0	0	0	0	0	97
	1962	0	0	87	0	0	0	0	86
	1978	0	83	0	0	0	0	0	85
	1976	0	85	0	0	0	0	0	83
	1969	0	73	0	0	0	0	0	78
	1990	0	71	30	50	0	0	10	71

Subcomponent decomposition provides redundancy across multiple mechanisms (full name, split-on-“and”, parenthetical extraction, embedded formal names, Public Law numbers). Co-occurrence rules target acts with generic names that require multi-term matching. Acts with zero total matches require investigation.

Matching conservatism for non-decomposable names

The subcomponent strategies only fire when specific structural patterns are present (“and”, parentheticals, “Act of YYYY”, Public Law numbers, event-type prefixes). For acts whose names lack these patterns, `generate_subcomponents` produces only the full name as a search term. Since Tier 2 matching uses `fixed()` substring search, the full string must appear contiguously in the chunk text.

Consider an act named “Internal Revenue Code of 1954”. None of the decomposition branches fire:

- No “ and ” to split on
- No parenthetical content
- The embedded formal name regex requires `Act of YYYY`, not `Code of YYYY`
- No Public Law number
- Does not start with Expiration/Suspension/Restoration/Changes

The only search term is the full name “`internal revenue code of 1954`”. A chunk that mentions “Internal Revenue Code” without the trailing “of 1954” would *not* match Tier 2. This is a recall gap: the matching is conservative for acts whose names do not decompose into shorter distinctive terms. The mechanism detail table above shows which acts rely on a single full-name term versus those with redundant subcomponent coverage.

Excluded act: Internal Revenue Code of 1954

The Internal Revenue Code of 1954 is excluded from evaluation data (`exclude_acts` in `align_labels_shocks()`). Its name doubles as the permanent legal name of the US tax code, so it is cited routinely across decades in contexts unrelated to the fiscal shock of its original enactment. Tier 2 matching flags every such mention as a positive, causing systematic contamination of evaluation data. Since the tier system is US-only infrastructure (Malaysia and other countries use the full codebook suite with expert validation), an act-level exclusion is the most defensible fix. The evaluation dataset contains 43 acts after this exclusion.

Temporal Consistency

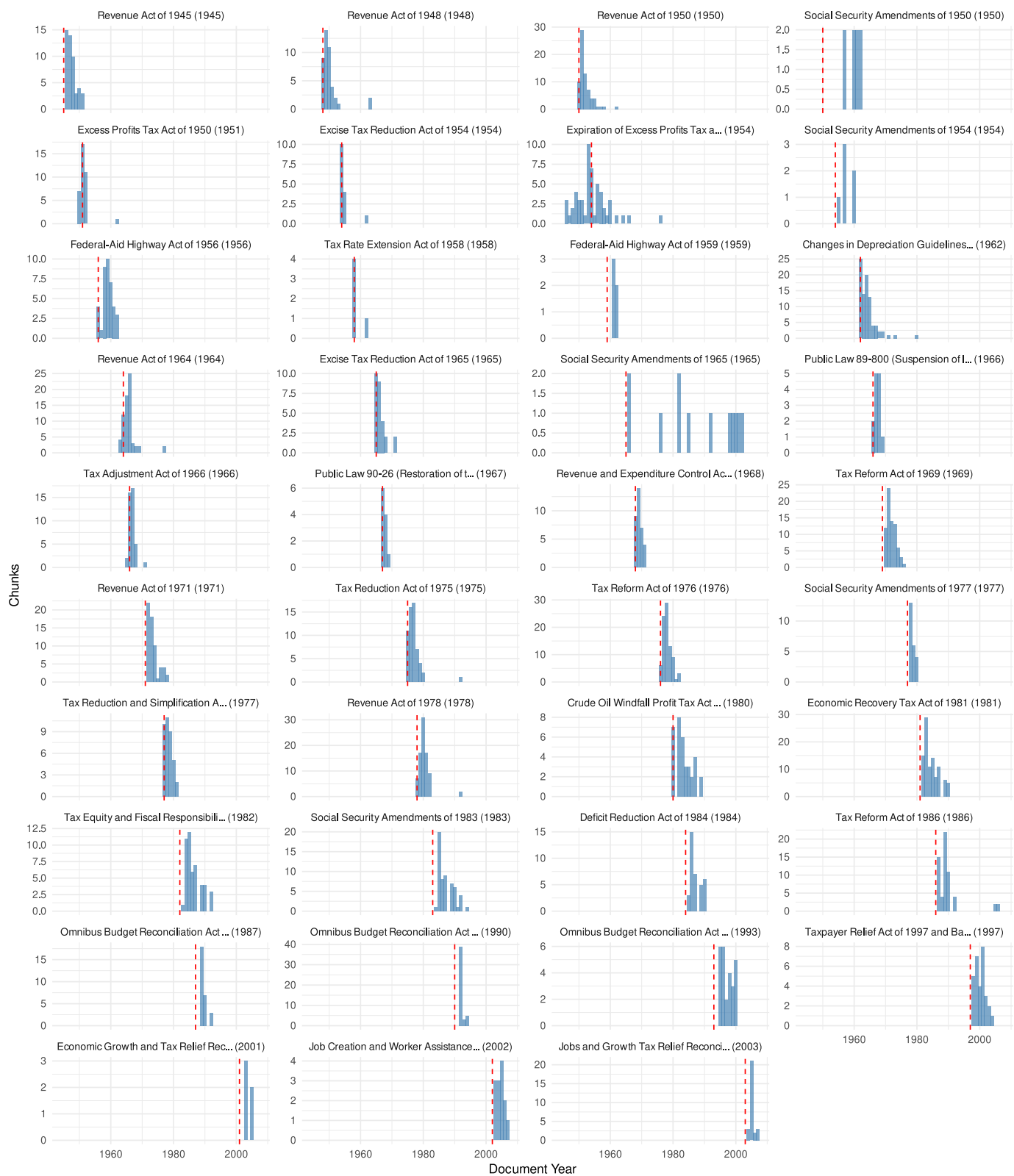


Figure 2: Document year distribution of Tier 2 matches per act (red line = signed year)

The red dashed line marks the act's signed year. Acts should show peak discussion around signing, with trailing references in subsequent years. Off-peak distributions suggest either: (1) anticipatory discussion (budget proposals), (2) retrospective analysis (ERPs), or (3) false positives from name collisions.

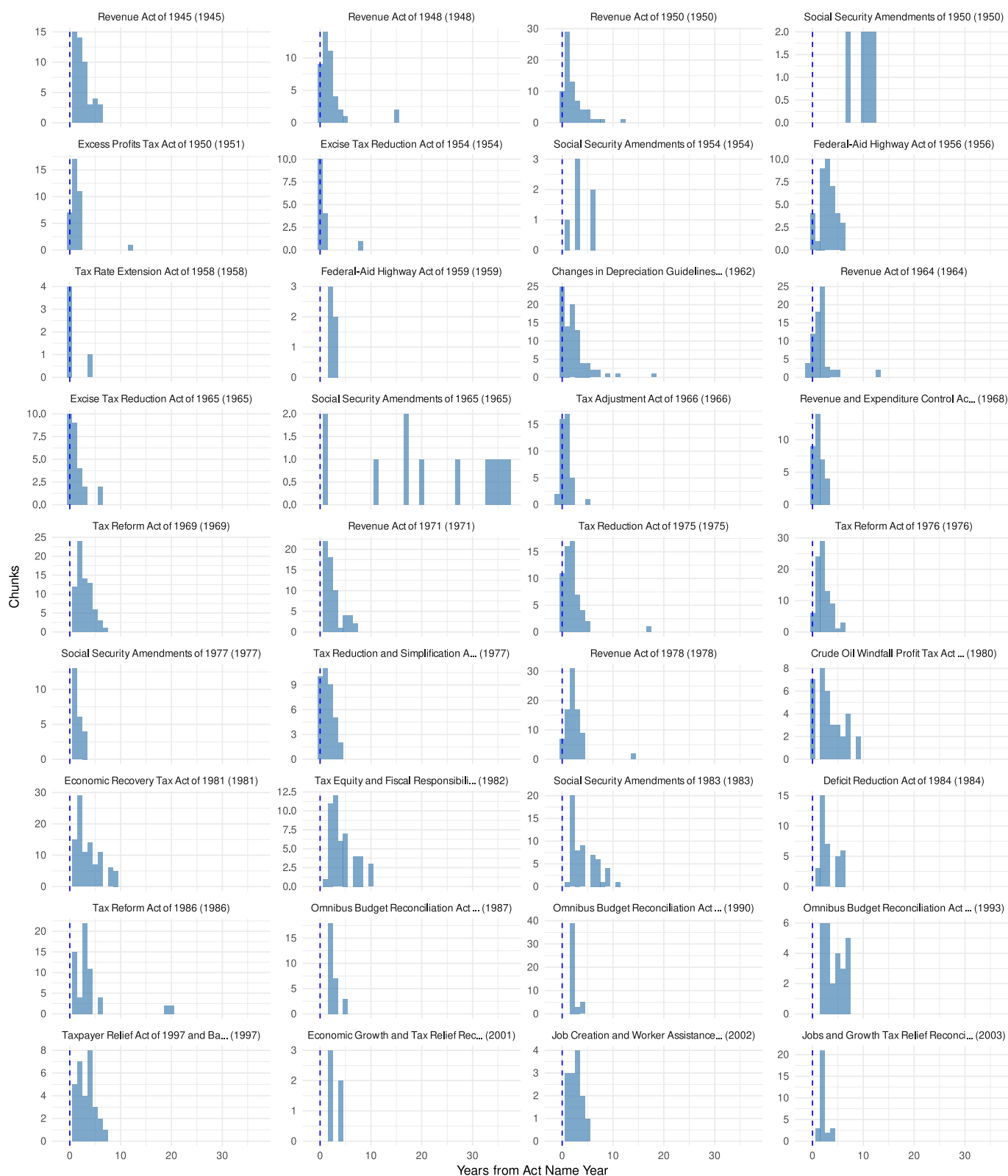


Figure 3: Year difference distribution for acts with embedded year (blue line = zero difference)

Table 6: Temporal precision for acts with embedded year

Act Name	Year	Chunks	Median Δ	$\pm 2\text{yr}$	$\pm 5\text{yr}$
Social Security Amendments of 1950	1950	8	10	0.0%	0.0%
Social Security Amendments of 1954	1954	6	3	16.7%	66.7%
Social Security Amendments of 1965	1965	12	24	16.7%	16.7%
Omnibus Budget Reconciliation Act of 1993	1993	26	4	23.1%	69.2%
Tax Equity and Fiscal Responsibility Act of 1982	1982	48	4	25.0%	77.1%
Tax Reform Act of 1986	1986	60	3	31.7%	86.7%
Federal-Aid Highway Act of 1956	1956	38	3	36.8%	92.1%
Social Security Amendments of 1983	1983	57	3	36.8%	66.7%
Taxpayer Relief Act of 1997	1997	30	3	40.0%	100.0%
Antiterrorism and Effective Death Penalty Act of 1996	1996	14	3	00.0%	100.0%
Small Business Job Protection Act of 1994	1994	14	3	00.0%	100.0%
Technical Corrections Act of 2002	2002	14	3	00.0%	100.0%
Antiterrorism and Effective Death Penalty Act of 1996	1996	14	3	00.0%	100.0%
Small Business Job Protection Act of 1994	1994	14	3	00.0%	100.0%
Technical Corrections Act of 2002	2002	14	3	00.0%	100.0%

Acts with low temporal precision (red rows, <25% within ± 2 years) likely suffer from name collisions or retrospective discussions dominating contemporaneous mentions. These require manual inspection to assess whether Tier 2 matches contain genuine positive examples.

Tier Overlap

Tier Overlap Test: PASS

Tier 1 and Tier 2 should be mutually exclusive by construction. Any overlap indicates a logic error in `identify_chunk_tiers()`. No overlap detected.

Token Distribution by Tier

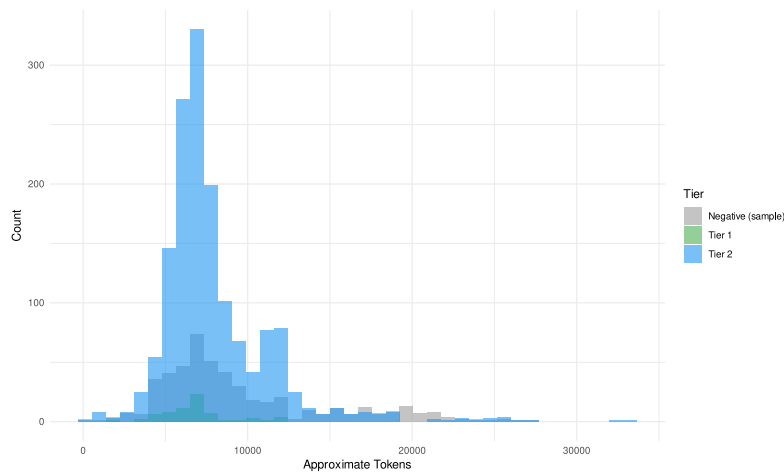


Figure 4: Chunk token distribution across tiers

Token distributions should be similar across tiers. Systematic differences would indicate selection bias (e.g., if Tier 1 only matches short documents). The negative sample should mirror positive tiers in distribution shape.

Key Density Distribution

The *key density* is a measure of how hard it is to correctly classify a negative chunk as positive. The key density measures the density of appearances of any matching keyword related to fiscal policy discussion over the total works in each chunk. Hence, a policy-heavy chunk should have a higher density key score. The keys considered were:

```

[1] "fiscal"          "tax"          "taxes"
"spending"
[5] "deficit"          "debt"         "gdp"
"unemployment"
[9] "stimulus"         "expenditure"  "revenue"
"excise"
[13] "appropriation"    "corporate"    "income"
"payroll"
[17] "tariff"           "levy"         "act"
"vat"

```

Again, because this step can only be performed for the US, we won't need to have a similar dictionary in other languages.

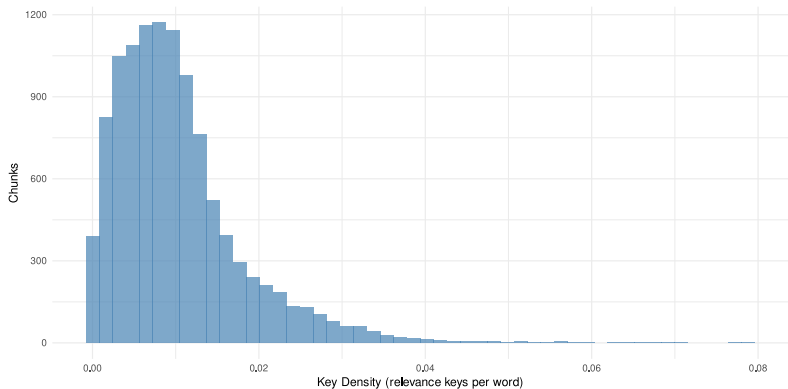


Figure 5: Key density distribution in negative pool

Table 7: Key density bin distribution in negative pool

Density Range	Chunks	Percentage
0	434	3.9%
(0, 1%]	6,078	54.5%
(1%, 3%]	4,345	38.9%
(3%, 5%]	265	2.4%
>5%	35	0.3%

The negative pool exhibits continuous `key_density` variation, enabling stratified precision analysis in S3. High-density negatives contain substantial fiscal vocabulary but lack explicit act mentions, providing challenging test cases where the model must distinguish general fiscal discussion from specific measure descriptions.

Negative Quality Check

Contamination Rate: 0.00% (0 / 11,157 chunks)

Table 8: Example contaminated negative chunks

status
No contaminated chunks detected

Table 9: Source type distribution in negative pool

Source Type	Chunks	Percentage
Budget	5,696	51.1%
Treasury	3,062	27.4%
ERP	2,399	21.5%

Low contamination rates (<1%) are acceptable, as these chunks likely discuss acts tangentially without containing identifiable passages. Source type balance ensures negatives represent the full document corpus, not biased subsets.

Rendered Tier 1 Examples

Visual inspection of Tier 1 matches confirms that verbatim passage text appears within the matched chunk. For each sample, we recover the passage from `aligned_data`, locate it in the chunk text, and highlight the match with surrounding context. Note that, we display text not as is in the source, but transformed for matching purposes.

Revenue Act of 1948

Doc: `AR_TREASURY-1948` | **Chunk:** 49 | **Passage:** 9

...e government revenues to such an extent asio make likely a deficit in government finances, at a time when responsible conduct of the financial affairs of this nation requires a substantial surplus in order to reduce our large public debt and to be reasonably prepared against contingencies.. second, **the bill would greatly increase the danger of further inflation, by adding billions of dollars of purchasing power** at a time when demand already exceeds supply at many strategic points in the economy, and when government expenditures are necessarily rising. the estimates of government expenditures for the fiscal year 1949 which i submitted to the congress in january totaled 39.7 billion dollars. r...

Excess Profits Tax Act of 1950

Doc: ERP-1950-Midyear | Chunk: 8 | Passage: 1

... should ensure that defense requirements and essential civilian needs have priority over other uses. hoarding must not be tolerated, nor can we allow needed scarce materials to be used for nonessential purposes. third, public policies should provide protection against the threat of a new inflation. **an increase of 10 billion dollars in military appropriations, coming at a time when our economy is** already operating at near-maximum levels, will greatly increase demand for goods and for labor, thus putting pressure on our price and wage structure. a sharp and continued rise in prices and wages at this time would not only impose hardships on domestic consumers; it would weaken our whole economy a...

Changes in Depreciation Guidelines and Revenue Act of 1962

Doc: ERP-1962 | Chunk: 5 | Passage: 4

...ce of these proposals is the 8 percent tax credit against tax for gross investment in depreciable machinery and equipment. the credit should be retroactive to january 1, 1962. the tax credit increases the profitability of productive investment by reducing the net cost of acquiring new equipment. **it will stimulate investment in capacity expansion and modernization, contribute to the growth of our** productivity and output, and increase the competitiveness of american exports in world markets. the tax credit for investment is in part self-financing. the stimulus it provides to new investment will have favorable effects on the level of economic activity during the year, and this will in turn add ...

Tax Adjustment Act of 1966

Doc: AR_TREASURY-1965 | Chunk: 3 | Passage: 1

...oted that we digitized for fraser <http://fraser.stlouisfed.org/> federal reserve bank of st. louis — page break — annual report on the financed xxiii have had a clear illustration of the direct relationship between tax policies, economic growth, and federal revenues. he went on to observe that: **“tax policy, however, must be used flexibly. we must be equally prepared to employ it in restraint of** an overly rapid economic expansion as we were to use it as a stimulus to a lagging economy. the current situation calls for a modest measure of fiscal restraint.” balance of payments the u.s.

balance-of-payments deficit increased sharply during the last half of calendar 1964, reaching an annual rat...

Federal-Aid Highway Act of 1959

Doc: ERP-1959 | **Chunk:** 9 | **Passage:** 4

...h success depends on a combination of appropriate public and private policies and actions, government must supply the required leadership and direction for its achievement. digitized for fraser <http://fraser.stlouisfed.org/> federal reserve bank of st. louis — page break — a balanced 1960 budget **the principal means by which government can express leadership in the effort to preserve price stability** is to conduct its own financial affairs prudently. the budget submitted to the congress for the fiscal year 1960, which balances expenditures with receipts at a level of \$77 billion, seeks to fulfill this responsibility. the expenditures estimated for fiscal 1960 are \$3.9 billion less than thos...

Rendered Tier 2 Examples

Tier 2 chunks match via act name subcomponents or co-occurrence rules. For each sample, we identify the matched term and show its location in the chunk text.

Economic Growth and Tax Relief Reconciliation Act of 2001

Doc: BUDGET-2005 | **Mechanism:** Full name | **Term:** economic growth and tax relief reconciliation act of 2001

...ation's currency through improved anti-counterfeiting measures. promoting america's prosperity surrounded by members of congress who supported the legislation, president bush signs egtrra on june 7, 2001. since 2001, the president has signed three tax bills to boost the economy. in combination, the **economic growth and tax relief reconciliation act of 2001** (egtrra) and the jobs and growth tax relief reconciliation act of 2003 (jgtrra) provided substantial tax relief and immediate stimulus to the economy, improving the potential for economic growth in the longrun. these acts reduced taxes by an average of \$1,544 for 109 million taxpayers in 2003. in a...

Revenue Act of 1950

Doc: AR_TREASURY-1950 | **Mechanism:** Full name | **Term:** revenue act of 1950

...n explanation of each increase or decrease during the year. taxation developments no major revenue legislation was enacted during the fiscal year 1950. a tax revision bill (h. r. 8920) was passed by the house on june 29, 1950, but developments in korea led to substantial changes in the bill and the **revenue act of 1950** was not approved finally until september 23, 1950. section i which follows summarizes the suggestions for the 1950 tax program made to the congress by the president and the secretary of the treasury and describes the major provisions of the revenue act of 1950. section ii summarizes the social secu...

Crude Oil Windfall Profit Tax Act of 1980

Doc: [AR_TREASURY-1980](#) | **Mechanism:** Full name | **Term:** [crude oil windfall profit tax act of 1980](#)

...tions will be made when a local government resubmits claims that may have been insufficiently reimbursed under the former regulations. • under public law 96-74, security for motorcades and at other places associated with the visit of a foreign dignitary may be reimbursed. tax policy legislation the **crude oil windfall profit tax act of 1980**.—the crude oil windfall profit tax act of 1980 (public law 96-223), proposed april 26, 1979, was enacted april 2, 1980. the tax applies generally to three categories or tiers of oil: oil previously subject to price controls; stripper oil; and newly digitized for fraser <http://fraser.stlouisfed.org/> fed...

Economic Recovery Tax Act of 1981

Doc: [BUDGET-1983](#) | **Mechanism:** Full name | **Term:** [economic recovery tax act of 1981](#)

...e the executive and legislative branches of our government joined together to respond to the mandate of the american people and overcome the impediments that had paralyzed washington for a decade. together, we have launched a process of reform and change that can transform the course of events. the **economic recovery tax act of 1981** is the largest, most comprehensive, and most constructive tax bill ever adopted. with the cooperation of the congress and support of the public, it was enacted in just 5 months. it addressed and substantially remedied most of the tax system's shortcomings and disincentives that had accumulated over...

Social Security Amendments of 1950

Doc: [AR_TREASURY-1961](#) | **Mechanism:** Full name | **Term:** [social security amendments of 1950](#)

...ed as estimated by the secretary of the treasury and adjusted in accordance with wage reports certified by the secretary of health, education, and welfare, for distribution to this fund and the federal disability insurance trust fund. 2 to cover employees of states and their political subdivisions, **social security amendments of 1950**. 3 excludes repayment of amortized premium and discount (net) amounting to \$23,019,401.41. * in connection with payments of benefits to survivors of certain world war ii veterans who died within three years after separation from active service, 6 incidental recoveries, etc., and, beginning with the...

Rendered High-Density Negatives

The top 3 negatives by [key_density](#) are the hardest negative cases. These chunks contain substantial fiscal vocabulary but no known act mentions. Inspecting them confirms they discuss general fiscal policy rather than specific enacted measures.

Negative 1

Doc: [AR_TREASURY-1980](#) | **Year:** 1980 | **Source:** Treasury | **Key density:** 0.0788

exhibits 453 capital formation taxes on capital income such as the corporate income tax and the individual income tax on interest and dividends reduce the after-tax return on savings. put another way, an income tax encourages present, as compared to future, consumption. with no taxes, a person with \$100 of income could choose between buying \$100 of consumption goods this year or saving now and buying \$110 of consumption goods next year, assuming the interest rate is 10 percent. thus, a person ca...

Negative 2

Doc: [AR_TREASURY-1958](#) | **Year:** 1958 | **Source:** Treasury | **Key density:** 0.0786

digitized for fraser ” <http://fraser.stlouisfed.org/> federal reserve bank of st. louis . 7 — page break — summary of fiscal operations net budget receipts were \$69.1 billion and budget expenditures were \$71.9 billion in the fiscal year 1958. to finance the deficit of \$2.8 billion and to increase the cash balance in the account of the treasurer of the united states, treasury borrowing during the year resulted in an increase of \$5.8 billion in the public debt. as of june 30, 1958, the public d...

Negative 3

Doc: AR_TREASURY-1980 | **Year:** 1980 | **Source:** Treasury | **Key density:** 0.0776

chart 2 output per hour, private nonfarm business sector os 140 120 100 80 ratio scale, index, 1967=100 1948 1948-1968 trend 1953 1958 1963 1968 1973 1978 digitized for fraser <http://fraser.stlouisfed.org/> federal reserve bank of st. louis — page break — charts cyclical comparisons of output per hour, private nonfarm business sector”^ index, peak quarter=100 : 110 ^05- 100 95 average of five previous cycles / current cycle / (1973-q4—1979-q2) j_j ± 1973 1975 *changes following the cyclical...

Summary and Conclusions

Table 10: Verification dashboard for chunk tier system

Verification Check	Result	Status
Coverage (all acts matched)	39 / 39 acts	PASS
Tier overlap (zero expected)	0 overlapping chunks	PASS
Temporal consistency (>50% acts within ±2yr)	23 / 36 acts	PASS
Negative contamination (<1%)	0.00%	PASS
Token distribution (similar across tiers)	Visual inspection	PASS

Interpretation

The chunk tier system provides a validated foundation for C1 (Measure ID) codebook evaluation. Key findings:

- **Positive identification** achieved for 39 / 39 acts through verbatim passage matches (Tier 1) and act name matching (Tier 2)
- **Matching mechanisms** show redundancy across full name, year-only, and co-occurrence strategies, improving recall
- **Temporal consistency** varies by act, with 64% of acts showing ≥50% matches within ±2 years of signing
- **Negative pool** spans 11,157 chunks with continuous key density distribution, enabling stratified hard-negative sampling

- **Minimal contamination** (0.00%) in negative pool confirms tier separation

Acts with WARN or FAIL status require manual inspection before use in LOOCV evaluation. See `identifying_known_acts.qmd` for the design rationale and `c1_measure_id.qmd` for Stage 2 evaluation using these tiers.

Next Steps

1. Investigate uncovered acts (if any) and assess whether missing coverage reflects data gaps or matching logic failures
2. Manually review contaminated negative chunks to determine if they represent true negatives or mislabeled positives
3. Use temporal consistency results to inform few-shot example selection (prioritize acts with high ± 2 yr precision)
4. Proceed to Stage 2 (Zero-Shot Evaluation) using validated tier assignments for LOOCV

Romer, Christina D., and David H. Romer. 2010. “The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks.” *American Economic Review* 100 (3): 763–801.